Q1 - 25 July - Shift 1

The letters of the word 'MANKIND' are written in all possible orders and arranged in serial order as in an English dictionary. Then the serial number of the word 'MANKIND' is _______.

Space for your notes:

Q2 - 26 July - Shift 1

The number of 5-digit natural numbers, such that the product of their digits is 36, is

Space for your notes:

Q3 - 26 July - Shift 2

Numbers are to be formed between 1000 and 3000, which are divisible by 4, using the digits 1,2,3,4,5 and 6 without repetition of digits. Then the total number of such numbers is ______________.

Space for your notes:

Q4 - 27 July - Shift 1

Let S be the sample space of all five digit numbers. If p is the probability that a randomly selected number from S, is a multiple of 7 but not divisible by 5, then $9p$ is equal to

(A) 1.0146 (B) 1.2085

(C) 1.0285 (D) 1.1521

Space for your notes:

Q5 - 28 July - Shift 1

Let S be the set of all passwords which are six to eight characters long, where each character is either an alphabet from {A, B, C, D, E} or a number from {1, 2, 3, 4, 5} with the repetition of characters allowed. If the number of passwords in S whose at least one character is a number from {1, 2, 3, 4, 5} is $\alpha \times 5^6$, then α is equal to _____.

Space for your notes:

Q6 - 28 July - Shift 2

A class contains b boys and g girls. If the number of ways of selecting 3 boys and 2 girls from the class is 168, then b + 3 g is equal to

Space for your notes:

Q7 - 29 July - Shift 1

Let $S = \{4, 6, 9\}$ and $T = \{9, 10, 11, \ldots, 1000\}$. If

$A = \{a_1 + a_2 + \ldots + a_k : k \in N, a_1, a_2, a_3, \ldots, a_k \in S\}$,

then the sum of all the elements in the set $T - A$ is equal to _______.

Space for your notes:

Q8 - 29 July - Shift 2

The number of natural numbers lying between 1012 and 23421 that can be formed using the digits 2, 3, 4, 5, 6 (repetition of digits is not allowed) and divisible by 55 is____,

Space for your notes:

Answer Key

Q1 (1492)	**Q2** (180)	**Q3** (30)	**Q4** (C)
Q5 (7073)	**Q6** (17)	**Q7** (11)	**Q8** (6)

Q1 (1492)

M	A	N	K	I	N	D

$$\left(\frac{4\times 6!}{2!}\right)+(5!\times 0)+\left(\frac{4!\times 3}{2!}\right)+(3!\times 2)+(2!\times 1)+(1!\times 1)+(0!\times 0)+1=1492$$

Q2 (180)

$$3\times\frac{5!}{2!2!}+\frac{5!}{3!\times 2!}+\frac{5!}{2!}+\frac{5!}{3!}=180$$

Q3 (30)

Here 1^{st} digit is 1 or 2 only

Case-I

If first digit is 1

Then last two digits can be 24, 32, 36, 52, 56, 64

1			

$1 \times 3 \times \underbrace{6}$ $= 18$ ways

Case – II

If first digit is 2 then last two digit can be 16, 36, 56, 64

2			

$1 \times 3 \times \underbrace{4}$ $= 12$ ways

Total ways = 12 + 18 = 30 ways

Q4 (C)

$n(S) = \text{all } 5 \text{ digit nos } = 9 \times 10^4$

A : no is multiple of 7 but not divisible by 5

Smallest 5 digit divisible by 7 is 10003

Largest 5 digit divisible by 7 is 99995

$\therefore$ 99995 = 10003 + (n – 1) 7 n = 12857

Numbers divisible by 35

99995 = 10010 + (P-1) 35 $\Rightarrow$ P = 2572

$\therefore$ Numbers divisible by 7 but not by 35 are

12857 – 2572 = 10285

$$\therefore P = \frac{10285}{90000} \qquad \therefore 9P = 1.0285$$

Ans. (C) [1.0285]

Q5 (7073)

Required no. = Total – no character from {1, 2, 3, 4, 5}

$= (10^6 - 5^6) + (10^7 - 5^7) + (10^8 - 5^8)$

$= 10^6 (1 + 10 + 100) - 5^6 (1 + 5 + 25)$

$= 10^6 \times 111 - 5^6 \times 31$

$= 2^6 \times 5^6 \times 111 - 5^6 \times 31$

$= 5^6 (2^6 \times 111 - 31)$

$= 5^6 \times \underbrace{7073}_{\alpha}$

$\therefore \alpha = 7073$

Q6 (17)

$$^{b}C_3 \times ^{g}C_2 = 168$$

$b(b-1)(b-2)(g)(g-1) = 8 \times 7 \times 6 \times 3 \times 2$

$b + 3g = 17$

Q7 (11)

$S=\{4,6,9\} \quad T = \{9,10,11 \ldots 1000\}$

$A\{a_1 + a_2 + \ldots + a_k : K \in N\} \ \& \ a_i \in S$

Here by the definition of set 'A'

$A = \{a : a = 4x + 6y + 9z\}$

Except the element 11, every element of set T is of of the form $4x + 6y + 9z$ for some $x, y, z \in W$

$\therefore T - A = \{11\}$

Q8 (6)

4 digit numbers

For divisibility by 55, no. should be div. by 5 and 11 both

			5
a	b	c	d

Also, for divisibility by 11

$a + c = b + 5$

for $b = 1$ $\quad a = 2, \quad c = 4$

$\quad a = 4, \quad c = 2$

for $b = 2$ $\quad a = 3, \quad c = 4$

$\quad a = 4, \quad c = 3$

for $b = 3$ $\quad a = 6, \quad c = 2$

$\quad a = 2, \quad c = 6$

$\therefore$ 6 possible four digit no.s are div. by 55

(II) 5 digit number is not possible

(Not possible)